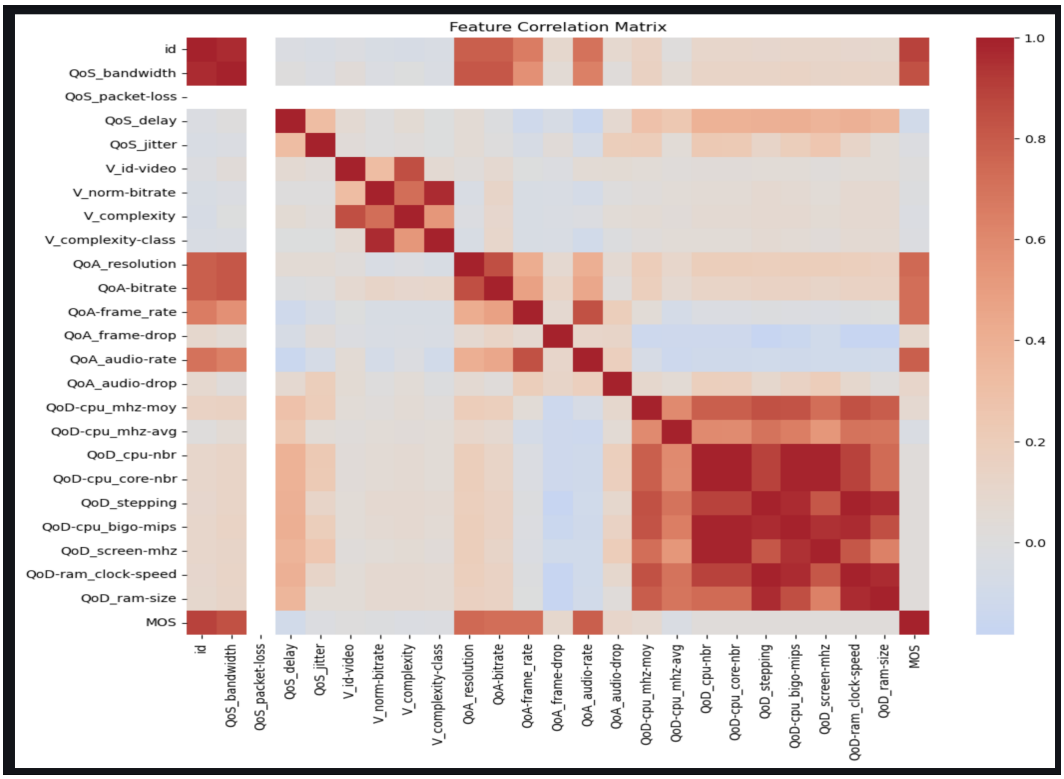
Exploratory Data Analysis Report: QoE Video Dataset

Feature Correlation Heatmap



This report summarizes the exploratory data analysis performed on the QoE Video Dataset that contains network parameters (QoS), video characteristics, playback indicators (QoA), device characteristics (QoD), and the user perceived quality represented by Mean Opinion Score (MOS). The dataset contains 300 observations and 30 attributes collected in a controlled laboratory environment to evaluate perceived video quality.

Dataset Overview: • Total Samples: 300 • Total Features: 30 • Target Variable: MOS (Mean Opinion Score) • MOS scale: 1 (Bad) → 5 (Excellent) Feature Categories: 1. QoS (Network conditions): bandwidth, delay, jitter, packet loss 2. Video attributes: bitrate, complexity, content type 3. QoA (Playback indicators from VLC): resolution, bitrate, frame rate, frame drops, audio metrics 4. QoD (Device characteristics): CPU specs, RAM, screen parameters

MOS Distribution: MOS values are reasonably distributed across the quality scale: 1: 38 samples 2: 56 samples 3: 77 samples 4: 65 samples 5: 64 samples The distribution indicates that the dataset covers a wide range of quality experiences, allowing meaningful analysis of the factors affecting perceived video quality.

QoS Analysis: Bandwidth: Bandwidth shows a strong positive relationship with MOS. Lower MOS values occur at bandwidth around 1 Mbps, while higher MOS values appear when bandwidth exceeds approximately 3 Mbps. This confirms that available network bandwidth is one of the most influential parameters for video streaming quality. Packet Loss: Packet loss remained constant at 0% across all experiments. As a result, packet loss does not contribute to MOS variation in this dataset and therefore cannot be used to model QoE effects. Delay: Network delay does not show a strong trend across MOS values. This suggests that buffering mechanisms in the video player likely absorb moderate delay variations, preventing users from perceiving latency differences. Jitter: Jitter values remain relatively stable across MOS levels, indicating that jitter variation was not significant enough to impact perceived video quality in these experiments.

Video Content Analysis: The dataset contains two content types: • Movie (204 samples) • Cartoon (96 samples) The MOS distributions for both categories are very similar. This indicates that content type does not significantly influence perceived quality under the tested conditions. Instead, network and playback parameters dominate QoE outcomes.

Resolution Impact on MOS: Video resolution shows a strong positive relationship with MOS: • 144p → MOS mainly between 1 and 2 • 240p → MOS around 2–3 • 360p → MOS around 3–4 • 480p → MOS around 4 • 720p and 1080p → MOS mostly above 4 This indicates that resolution is a major factor influencing visual quality perception. Higher resolutions provide clearer and more detailed video frames, leading to higher user satisfaction.

Correlation Analysis: The correlation heatmap highlights several important relationships: Strong positive correlations with MOS: • QoS_bandwidth • QoA_resolution • QoA_frame_rate • QoA_bitrate • QoA_audio_rate Weak correlations: • Network delay • Jitter • Video complexity • Device characteristics Highly correlated device features (CPU, RAM, etc.) indicate multicollinearity among hardware attributes, suggesting that device parameters may not independently contribute to QoE in this dataset.

Key Findings: 1. Network bandwidth is the strongest QoS driver of perceived video quality. 2. Video playback indicators such as resolution, frame rate, and bitrate strongly affect MOS. 3. Content type (movie vs cartoon) does not significantly impact QoE. 4. Packet loss was not varied in the experiments and therefore has no influence in the dataset. 5. Device characteristics appear less influential compared to network and playback metrics. 6. Higher resolution and bitrate combinations consistently correspond to better user experience.

Future Analysis Directions: The insights obtained from EDA provide a strong foundation for building QoE prediction models. Future work may include: • Feature preprocessing and normalization • Machine learning models for MOS prediction • Feature importance analysis using Random Forest or Gradient Boosting • Explainable AI techniques such as SHAP to understand QoE drivers • Comparative analysis between QoS, QoA, and QoD influence factors